

In fact Microsoft themselves offers a tool to mount iso files since they offer a lot of their developer focused software through iso files. This feature is even inbuilt in Windows 8 and above where you can just double click on an iso file to have it show up as a virtual drive. Eject it, and poof, it's gone.

Now that we have eased you in with the concept of emulating a single device, let us assure you, almost any device you can think of, chances are there is an emulator for it, and for good reason.

Optical disk emulation

We've already gone on about how optical disk drives can be emulated, but there is more to them than we have covered here. While Microsoft's official optical disk emulation software (Virtual CD-ROM Control Panel) is useful, it only works with iso images, and there are a lot many more formats out there. If you want to deal with them, you will need a better optical drive emulation software. Let's also look at why there are so many formats for storing the same kind of information, why doesn't everyone just use .iso files? The answer is that is DRM.

Back before online activation became ubiquitous, a common way to prevent piracy was to require that the user keep the disk that came with the software inserted in the drive while the software was running — or in some cases when the software launched or at random times when asked. The presence of this disk would signal to the software that it was legally purchased, this forced a single computer to run the software at any given time.

With the availability of cheap CD burners this became a problem. People could simply copy a disk for cheap and use it on multiple computers. How do you counter this? Make it harder to duplicate CDs of course! So software publishers began using specially manufactured disks that would have data stored in special locations that couldn't be duplicated on a writable disk. Others would have deliberate errors on special locations on the disk. These errors would prevent copying using standard software, or at least make it impossible to replicate the disk exactly.

This is where specialised optical disk emulation software came in. These software, such as Daemon tools and Alcohol 120% knew these tricks and could store information about these special tricks in the image file. This requires specialised formats for storing images as .iso files cannot store this information. Also, it required special software that could emulate these tricks and fool the game or software in question.